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Topic: RHEL 8 LVM Installation, Disk Expansion, Logical Volume Creation, and Persistent Mounting

1. Boot up a RHEL 8 virtual machine and perform a custom installation using the graphical installer, making sure to select 'LVM' as the partitioning scheme for the root (/) partition.

⇒ Boot the RHEL 8 Virtual Machine using the ISO image.

Select:

Install Red Hat Enterprise Linux 8

Choose language and keyboard settings.

Go to:

Installation Destination

Select the virtual disk.

Under Storage Configuration choose:

Custom

Create partitions:

Example:

Mount Point	Size	Type
/boot	1 GB	Standard Partition
/	Remaining Space	LVM

Create LVM Volume Group:

Example:

VG Name: rhel

Create Logical Volume:

Example:

LV Name: root

Mount Point: /

Filesystem: xfs

Click:

Done → Accept Changes → Begin Installation

After installation, reboot into RHEL 8.

2. Inside your RHEL 8 system, use the 'lsblk' and 'vgs' commands to verify that your root partition is managed by LVM and note down the Volume Group and Logical Volume names.

⇒ Check block devices:

lsblk

The root partition is:

/dev/mapper/rhel-root

which shows that / is inside LVM.

Check Volume Groups:

vgs

3. Add a new virtual disk to your RHEL 8 VM, then use 'pvcreate', 'vgextend', and 'lvextend' to increase the size of your existing root Logical Volume by at least 1GB, and resize the filesystem to use the new space.

Hint: Use 'df -h' before and after to confirm the size change.

⇒ Step 1: Add a new virtual disk

In VM settings:

Add Storage → Create New Virtual Disk → 2GB

Boot RHEL 8.

Check new disk:

lsblk

Create PV:

sudo pvcreate /dev/sdb

Verify:

Pvs

Add new PV to existing VG:

```
sudo vgextend rhel /dev/sdb
```

Verify:

```
Vgs
```

Check current root size

```
df -h /
```

Resize filesystem

For RHEL 8 default filesystem (XFS):

```
sudo xfs_growfs /
```

If filesystem is ext4:

```
sudo resize2fs /dev/rhel/root
```

Confirm size increase

```
df -h /
```

4. Create a new Logical Volume called 'musicdata' of 500MB in your existing Volume Group, format it with ext4, mount it at /mnt/music, and add an entry to /etc/fstab so it mounts automatically on boot.

Constraint: Use only command-line tools (no GUI) for this task.

⇒ Create 500MB LV:

```
sudo lvcreate -L 500M -n musicdata rhel
```

Verify:

```
Lvs
```

Format with ext4

```
sudo mkfs.ext4 /dev/rhel/musicdata
```

Step 3: Create mount point

```
sudo mkdir /mnt/music
```

Step 4: Mount Logical Volume

```
sudo mount /dev/rhel/musicdata /mnt/music
```

Verify:

```
df -h
```

Step 5: Add entry in /etc/fstab

Find UUID:

```
sudo blkid /dev/rhel/musicdata
```

Edit fstab:

```
sudo vi /etc/fstab
```

Add:

```
UUID=abcd-1234 /mnt/music ext4 defaults 0 0
```

Save and exit.